The deflation correction as implemented

The formulas behind `lgh_glr_correct` / `lgh_glr_correct_probes` (stage 2.5 of `lgpsf-hessian`).
Companions: `glr-woodbury-formulas.pdf` (the base representation), `glr-distributed-design.pdf` (layouts).

Definitions: M lumped mass (diagonal); Z the prior's square-root factor, prior precision $R = ZM^{-1}Z$; H_d the *true* misfit Hessian (matvecs only); $\widetilde{H}_d$ the sparse lgpsf approximation; the GLR build holds $U\Lambda U^T \approx F := M^{1/2}Z^{-1}\widetilde{H}_d Z^{-1}M^{1/2}$ (treated, truncated, $U^TU = I_r$). The approximated and true operators at shift c:

$$\widetilde{H}(c) = ZM^{-1/2}\,(U\Lambda U^T + cI)\,M^{-1/2}Z, \quad H(c) = H_d + cR = ZM^{-1/2}\,(F_{\text{true}} + cI)\,M^{-1/2}Z, \quad (1)$$

with $F_{\text{true}} := M^{1/2}Z^{-1}H_dZ^{-1}M^{1/2}$ (never formed; one true apply per column).

1. The error eigenvalue problem

At a reference shift $c_0 > 0$, the correction targets the dominant eigenpairs of the generalized problem

$$\big(H(c_0) - \widetilde{H}(c_0)\big)\,v \;=\; d\,\widetilde{H}(c_0)\,v, \qquad H(c_0) - \widetilde{H}(c_0) = H_d - \widetilde{H}_d^{\text{glr}} \;\; \text{(the regularization cancels).} \quad (2)$$

In prior-whitened coordinates, with $S^{\pm} := (U\Lambda U^T + c_0I)^{\pm 1/2}$ (one filter apply each via the master formula, $S^{\pm} = U\big[(\Lambda + c_0I)^{\pm1/2} - c_0^{\pm1/2}I\big]U^T + c_0^{\pm1/2}I$, i.e. $\texttt{lgh\_glr\_pow}(c_0, \pm\frac{1}{2})$), (2) is the ordinary symmetric problem for the whitened error operator

$$E \;=\; S^{-}\,(F_{\text{true}} + c_0I)\,S^{-} \;-\; I \;=\; S^{-}\left(F_{\text{true}} - U\Lambda U^T\right)S^{-}. \quad (3)$$

Facts used by the implementation: (i) the eigenvalues d of E are those of (2) (congruence); (ii) $\operatorname{eig}\big(\widetilde{H}^{-1}H\big) = 1 + d$; (iii) if $H(c_0)$ is PD then $1 + d > 0$ exactly, so any Rayleigh estimate below $-1 + \varepsilon$ is estimation noise and is clamped there (`clamp_eps`, default 0.05; clamps are counted, not hidden).

2. The correction and its c-independence

With (V, D) the kept eigenpairs of E ($V^TV = I$, D diagonal, $|d|$-descending, clamped),

$$F_{\text{true}} \;\approx\; U\Lambda U^T + \widehat{V}D\widehat{V}^T, \qquad \widehat{V} := S^{+}V, \quad (4)$$

and hence, for *every* shift c (the stored object approximates the c-free error $F_{\text{true}} - U\Lambda U^T$; c_0 enters only as the metric selecting which modes dominate),

$$H(c) \;\approx\; ZM^{-1/2}\big(U\Lambda U^T + \widehat{V}D\widehat{V}^T + cI\big)M^{-1/2}Z. \quad (5)$$

3. Value pass (probe reuse; `lgh_glr_correct_probes`)

Given the fit stage's probe pairs ($\Omega,\ Y = H_d\Omega$), k_f columns after holding out the trailing n_{qc}:

$$X = M^{-1/2}Z^T\Omega \quad \text{(cheap applies)}, \qquad F_{\text{true}}X = M^{1/2}Z^{-1}Y \quad \text{(blocked } Z \text{ solves)}, \quad (6)$$

$$R_w = F_{\text{true}}X - U\Lambda U^TX, \qquad E\,(S^{+}X) = S^{-}R_w \quad \text{(zero new true applies).} \quad (7)$$

An *energy-ordered* orthonormal basis of the free images: eigendecompose the Gram matrix $G = (S^- R_w)^T (S^- R_w) = W \Sigma W^T$, drop $\sigma_i \le \texttt{drop\_tol} \cdot \sigma_{\max}$, set $Q = (S^- R_w)\, W_+ \Sigma_+^{-1/2}$ (columns in descending energy). The apply budget m then prices directions exactly:

$$T \;=\; \mathrm{sym}\big(Q_m^T\, E\, Q_m\big), \qquad EQ_m = S^-\big((F_{\text{true}} + c_0 I)(S^- Q_m)\big) - Q_m \quad (m \text{ true applies}), \tag{8}$$

$T \;=\; V_s D V_s^T$, $V \;=\; Q_m V_s$. If budget remains beyond the basis dimension, one power step: orthogonalize EQ_m against Q_m, take the energy basis of the remainder, price it with the remaining applies, and assemble T on the combined block. Kept modes are $|d|$-descending, clamped (§1), and truncated at `d_keep_min`: values priced below the estimation-noise level add noise rather than information, and if no mode clears the threshold the object is left uncorrected (the report still carries the observed d window).

4. Fresh sketch (`lgh_glr_correct`)

Without probe data: a hashed Gaussian test block Ω_E ($m_r = \lfloor m/(2+q) \rfloor$ columns), images $E\,\Omega_E$ (m_r true applies per pass, q optional power passes through the energy basis), then the same Rayleigh pricing on the final basis — $\approx 2m_r$ applies for a rank-m_r correction.

5. Graft

The correction is folded into the representation, after which every downstream operation (solve, apply, sample, factor, logdet, pow, filter) serves the corrected operator through the unchanged master formula:

$$C_U = U^T \widehat{V}, \qquad Q_V = \mathrm{orth}_{\text{energy}}\big((I - UU^T)\widehat{V}\big), \qquad C_Q = Q_V^T \widehat{V}, \tag{9}$$

$$T_2 = \begin{pmatrix} \Lambda + C_U D C_U^T & C_U D C_Q^T \\ C_Q D C_U^T & C_Q D C_Q^T \end{pmatrix} = W_2 \Lambda' W_2^T \qquad (\text{dense symmetric, } r + r_2), \tag{10}$$

$$U' = [\,U \;\; Q_V\,]\, W_2 \quad (\text{truncated by } |\lambda'| \text{ with the object's rules; } \textbf{signed} \text{ — no treatment}). \tag{11}$$

Validity of shift-taking operations becomes the exact scalar condition

$$c \;>\; \text{floor} \;=\; \max(0,\; -\min_i \lambda_i') \qquad (\text{reported; floor} < c_0 \text{ guaranteed by the clamp}). \tag{12}$$

After a graft the sketch no longer represents the operator: `lgh_glr_extend` is refused (order: compute → extend → correct last), `lgh_glr_apply` switches from the exact sparse action to the corrected low-rank action, and ScaLAPACK-built objects first materialize the implicit $U = QV_{\text{kept}}$ (panel-wise) and drop the sketch state.

6. Cost summary

Per column of width-m blocks; "true apply" means one H_d matvec (typically a forward + adjoint PDE solve).

	true applies	blocked Z solves	dense work
value-pass basis	0	$O(k_f)$	Gram eig k_f
value-pass pricing (rank m)	m	$O(m)$	eig m
fresh sketch (rank m_r)	$(2{+}q)\, m_r$	$O((2{+}q)m_r)$	eig m_r
graft	0	0	eig $r + r_2$